TECHNICAL REPORT

North-West Electrical Engineering Consultants

Network Design and Implementation

Assigned feature: Access Control Lists (ACLs)

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1. Introduction

This report presents the design, implementation and testing of a secure and scalable network for North-West Electrical Engineering Consultants, an engineering company based in Klerksdorp that provides electrical consulting services to clients across the region.

The company needs a reliable network to support its daily work. This includes access to engineering design software, sharing of large technical documents, client communication, boardroom presentations, and administrative functions such as payroll and HR record management.

The project was completed across three stages:

- **Milestone 1 - Client design review:** requirements, topologies and the IP addressing plan.
- **Milestone 2 - Implementation review:** VLANs, ACLs and testing evidence.
- **Final submission:** full documentation and demonstration.

The networking feature assigned for this project was **Access Control Lists (ACLs)**. They were implemented on a Cisco 3560 Layer 3 core switch to enforce strict security policies between departments.

2. Client Requirements Analysis

2.1 Client Overview

North-West Electrical Engineering Consultants requires a stable, secure and scalable network that supports current operations and allows for future growth.

2.2 Functional Requirements

- **Network connectivity:** all staff must be able to connect their computers and devices to the network, and different departments must be able to share information when needed.
- **Boardroom support:** the boardroom must have both wired Ethernet ports and Wi-Fi access for presentations.
- **IP address planning:** the assigned block 192.168.34.0/24 must be divided into logical subnets with room for expansion.
- **Security (ACLs):** ACLs must filter traffic between departments, blocking unauthorised access and permitting the required traffic flows.
- **Remote management (CR9):** one off-site administrator must be able to manage network devices securely using SSH, not Telnet.
- **Documentation:** all work must be documented in GitHub and in a technical report.

2.3 Boardroom Constraint

The boardroom needs wired ports for stable, high-speed presentation connections and wireless access for visitors using laptops, tablets or phones. Both must work at the same time. To meet this, all boardroom devices were placed in VLAN 30, which keeps them separate from the internal departments and protects sensitive company data.

2.4 ACL Challenge Overview

ACLs act like security guards at the doorway of each department. They decide who is allowed in, who is denied, and what type of traffic may pass.

3. Network Design

3.1 Design Approach

A switch-based design was chosen instead of a traditional router-based design for four reasons:

- **Speed:** Layer 3 switches route traffic in hardware, while routers process traffic in software.
- **Scalability:** additional switches can be added as the company grows.
- **Cost:** a single Layer 3 switch performs both the switching and the routing functions.
- **Modern practice:** this matches how real enterprise networks are designed.

3.2 Physical Topology

A Cisco 3560 Layer 3 switch (SW-CORE) in the server room connects to one access switch for each area. The devices are placed as follows:

- **Server room:** SW-CORE and Server-1 (file server, VLAN 40).
- **Engineering:** SW-ENG with workstations PC-ENG-01 to PC-ENG-20 (VLAN 10).
- **HR and Admin:** SW-HR with workstations PC-HR-01 to PC-HR-15 (VLAN 20).
- **Boardroom:** SW-BOARD with two presentation PCs (PC-BOARD-1 and 2) and the wireless access point WAP-BOARD (VLAN 30).
- **Management:** SW-MGMT with workstations PC-MGMT-01 to PC-MGMT-03 (VLAN 50).
- **Off-site:** the remote administrator's management PC (VLAN 99).

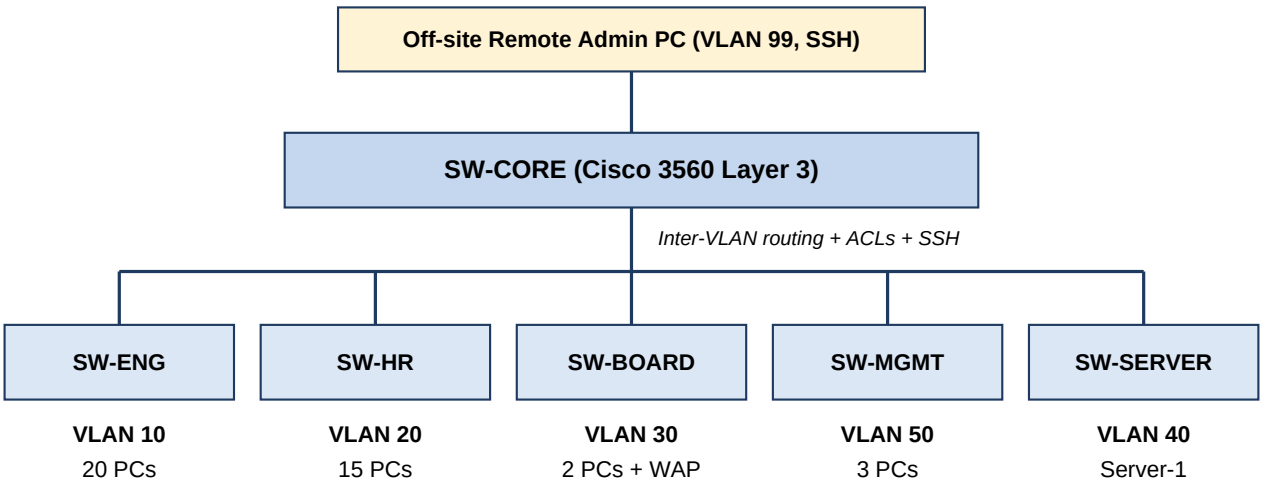


Figure 1: Physical and logical topology of the network

3.3 Logical Topology

Logically, SW-CORE is the single point where inter-VLAN routing, ACL filtering and SSH management take place. Each VLAN is a separate subnet with its own gateway on a Switch Virtual Interface (SVI) on the core switch. Traffic between VLANs must pass through the core, so every ACL is enforced at that one point.

3.4 VLAN Design

Six VLANs were created:

- **VLAN 10 - Engineering:** engineering workstations.
- **VLAN 20 - HR-Admin:** HR and administrative staff.
- **VLAN 30 - Boardroom:** presentation devices and the wireless access point.
- **VLAN 40 - Servers:** file server infrastructure.
- **VLAN 50 - Management:** network device management.
- **VLAN 99 - Native / Remote Admin:** remote administration and untagged traffic.

3.5 Trunk Configuration

802.1Q trunk links were configured between all access switches and the core switch, using native VLAN 99. This isolates untagged traffic and provides a secure management path.

3.6 DHCP Assignment

The core switch acts as the DHCP server for VLANs 10, 20, 30, 40 and 50, handing out addresses from each VLAN's own subnet. The default gateway in each case is the core switch's SVI address for that VLAN (listed in section 4.1). VLAN 99 uses static addressing because it only contains the remote administrator's PC.

4. IP Addressing Plan

Starting from the assigned block 192.168.34.0/24, Variable Length Subnet Masking (VLSM) was used to divide the network. Each subnet is sized for its department with room to grow, and unused address space remains for future VLANs.

4.1 Subnet Allocation

VLAN	Name	Network	Usable range	Broadcast	Gateway (SW-CORE)
10	Engineering	192.168.34.0/27	.1 - .30	.31	192.168.34.1
20	HR-Admin	192.168.34.32/27	.33 - .62	.63	192.168.34.33
30	Boardroom	192.168.34.64/28	.65 - .78	.79	192.168.34.65
40	Servers	192.168.34.80/28	.81 - .94	.95	192.168.34.81
50	Management	192.168.34.96/28	.97 - .110	.111	192.168.34.97
99	Remote Admin	192.168.34.128/28	.129 - .142	.143	192.168.34.129

4.2 Subnet Sizing Justification

- **Engineering (/27):** 20 engineers today, leaving 10 spare addresses for growth.
- **HR-Admin (/27):** 15 staff today, leaving 15 spare addresses for growth.
- **Boardroom (/28):** 4 devices today, which is enough for future presentations.
- **Servers (/28):** 1 server today, with room for additional servers.
- **Management (/28):** 3 devices today, with room for growth.
- **Remote Admin (/28):** a single administrator, with addresses reserved for future administrators.

4.3 Static Device Addresses

Most devices receive their addresses by DHCP. The following devices use fixed addresses:

- **WAP-BOARD (VLAN 30):** 192.168.34.73, gateway 192.168.34.65.
- **Server-1 (VLAN 40):** 192.168.34.90, gateway 192.168.34.81.
- **SW-CORE management interface (VLAN 50):** 192.168.34.100, gateway 192.168.34.97.
- **Remote Admin PC (VLAN 99):** 192.168.34.130, gateway 192.168.34.129.

5. Implementation

5.1 Inter-VLAN Routing

The Cisco 3560 Layer 3 switch performs inter-VLAN routing through Switch Virtual Interfaces (SVIs). IP routing was enabled with the `ip routing` command. The output of `show ip route` lists all six subnets as directly connected (C entries), which confirms that the core switch can route between every VLAN.

5.2 Assigned Feature - Access Control Lists (ACLs)

Four extended ACLs were implemented on the core switch. Extended ACLs were used because they can filter on source address, destination address and protocol, which is what the department policies require.

ACL 1: ACL_ENG_HR - block Engineering from HR

```
ip access-list extended ACL_ENG_HR
 deny ip 192.168.34.0 0.0.0.31 192.168.34.32 0.0.0.31
 permit ip any any
```

Applied to: VLAN 10, inbound. **Purpose:** prevents Engineering staff from accessing sensitive HR and payroll data.

ACL 2: ACL_BOARD_FINAL - block Boardroom from internal departments

```
ip access-list extended ACL_BOARD_FINAL
 deny ip 192.168.34.64 0.0.0.15 192.168.34.0 0.0.0.31
 deny ip 192.168.34.64 0.0.0.15 192.168.34.32 0.0.0.31
 deny ip 192.168.34.64 0.0.0.15 192.168.34.96 0.0.0.15
 permit ip any any
```

Applied to: VLAN 30, inbound. **Purpose:** visitors in the boardroom cannot reach Engineering, HR or Management, but can still reach the shared server and the internet.

ACL 3: ACL_ADMIN_SSH - restrict SSH to the remote administrator

```
ip access-list extended ACL_ADMIN_SSH
 permit tcp host 192.168.34.130 any eq 22
 deny tcp any any eq 22
 permit ip any any
```

Applied to: VLAN 99, inbound. **Purpose:** only the off-site administrator (192.168.34.130) may open SSH sessions to network devices.

ACL 4: ACL_ENG_SSH_BLOCK - secondary SSH protection

```
ip access-list extended ACL_ENG_SSH_BLOCK
 deny tcp 192.168.34.0 0.0.0.31 host 192.168.34.129 eq 22
 permit ip any any
```

Applied to: VLAN 10, inbound. **Purpose:** stops Engineering devices from reaching the SSH service on the core switch.

5.3 SSH Configuration

SSH version 2 was configured for remote management, and VTY access was restricted with a standard ACL so that only the administrator's address can connect:

```
ip domain-name project.local
ip ssh version 2
username admin privilege 15 secret cisco123
access-list 10 permit host 192.168.34.130
line vty 0 4
 access-class 10 in
 login local
 transport input ssh
```

The command `show ip ssh` confirms that SSH version 2.0 is enabled with RSA keys. Because `transport input ssh` is set, Telnet is not accepted.

5.4 ACL Application Summary

All ACLs are applied inbound on the SVI of the VLAN they protect, so unwanted traffic is dropped as close to its source as possible:

- **VLAN 10 (Engineering):** ACL_ENG_HR and ACL_ENG_SSH_BLOCK
- **VLAN 30 (Boardroom):** ACL_BOARD_FINAL
- **VLAN 99 (Remote Admin):** ACL_ADMIN_SSH

5.5 Trunk Configuration

The trunk configuration below was applied to ports Fa0/1 to Fa0/3 and Gig0/1 to Gig0/2:

```
interface FastEthernet0/1
 switchport trunk native vlan 99
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

5.6 Remote Admin Port

Port Fa0/4 on the core switch connects the remote administrator's PC and is placed in VLAN 99:

```
interface FastEthernet0/4
 switchport access vlan 99
 switchport mode access
```

6. Testing and Verification

All tests were performed in Cisco Packet Tracer in Realtime mode, after the network had fully converged. Six tests covered both the traffic that must be allowed and the traffic that must be blocked.

6.1 Test Results Summary

#	Test	Expected	Actual result	Status
1	Engineering PC to Server (ping 192.168.34.90)	Success	4 replies, 0% loss	**Passed**
2	Engineering PC to HR PC (ping 192.168.34.40)	Blocked	Destination unreachable, 100% loss	**Passed**
3	Boardroom PC to HR PC (ping 192.168.34.40)	Blocked	Destination unreachable, 100% loss	**Passed**
4	Boardroom PC to Server (ping 192.168.34.90)	Success	4 replies, 0% loss	**Passed**
5	Remote Admin SSH (ssh -l admin 192.168.34.129)	Success	Logged in as SW-CORE#	**Passed**
6	Engineering SSH (ssh -l admin 192.168.34.129)	Blocked	Connection refused	**Passed**

6.2 Test 1: Inter-VLAN Routing (Allowed)

```
C:\> ping 192.168.34.90
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Reply from 192.168.34.90: bytes=32 time=7ms TTL=127
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Analysis: a TTL of 127 shows that the packets crossed exactly one routed hop, the core switch. Engineering can therefore reach the shared server.

6.3 Test 2: ACL_ENG_HR (Blocked)

```
C:\> ping 192.168.34.40
Reply from 192.168.34.1: Destination host unreachable.
Reply from 192.168.34.1: Destination host unreachable.
Reply from 192.168.34.1: Destination host unreachable.
Reply from 192.168.34.1: Destination host unreachable.
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Analysis: the core switch (192.168.34.1) reports the destination as unreachable. This shows that ACL_ENG_HR drops the traffic at the VLAN 10 SVI.

6.4 Test 3: ACL_BOARD_FINAL (Blocked)

```
C:\> ping 192.168.34.40
Reply from 192.168.34.65: Destination host unreachable.
Reply from 192.168.34.65: Destination host unreachable.
Reply from 192.168.34.65: Destination host unreachable.
Reply from 192.168.34.65: Destination host unreachable.
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Analysis: the boardroom gateway (192.168.34.65) blocks the traffic to HR, which shows that ACL_BOARD_FINAL is active.

6.5 Test 4: Boardroom to Server (Allowed)

```
C:\> ping 192.168.34.90
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Reply from 192.168.34.90: bytes=32 time<1ms TTL=127
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Analysis: the final permit rule in ACL_BOARD_FINAL allows access to the server while the earlier deny rules block the internal departments.

6.6 Test 5: Remote Admin SSH (Allowed)

```
C:\> ssh -l admin 192.168.34.129
Password: ****
SW-CORE#
```

Analysis: the SSH version 2 login succeeds from the authorised address, so Change Request 9 is satisfied.

6.7 Test 6: Engineering SSH (Blocked)

```
C:\> ssh -l admin 192.168.34.129
% Connection refused by remote host
```

Analysis: only the authorised administrator can open an SSH session, and Telnet is disabled.

6.8 ACL Match Counters

The command `show ip access-lists` showed live match counters, which prove that the ACLs were actively enforcing the policy during testing:

```
Extended IP access list ACL_ENG_HR
    10 deny ip 192.168.34.0 0.0.0.31 192.168.34.32 0.0.0.31 (4 match(es))
    20 permit ip any any (4 match(es))

Extended IP access list ACL_ADMIN_SSH
    10 permit tcp host 192.168.34.130 any eq 22 (66 match(es))

Extended IP access list ACL_BOARD_FINAL
    20 deny ip 192.168.34.64 0.0.0.15 192.168.34.32 0.0.0.31 (4 match(es))
    40 permit ip any any (4 match(es))

Standard IP access list 10
    10 permit host 192.168.34.130 (10 match(es))
```

These counters show that unauthorised traffic was blocked and authorised traffic was permitted, as designed.

7. Conclusion

The project delivered a secure, scalable and fully functional network for North-West Electrical Engineering Consultants.

7.1 Requirements Met

Every client requirement was completed and verified: all staff can connect, departments can communicate under ACL control, the boardroom has both wired and wireless access, the IP plan uses VLSM, the ACLs (the assigned feature) are in place, remote SSH management (CR9) works, and the work is documented in GitHub and in this report.

7.2 Key Achievements

- Designed a hierarchical network around a Cisco 3560 Layer 3 core switch.
- Segmented the network into six VLANs for security and performance.
- Configured inter-VLAN routing using SVIs.
- Implemented four extended ACLs that enforce the departmental security policies.
- Configured SSH version 2 with RSA keys and restricted VTY access with standard ACL 10.
- Verified all security and connectivity requirements through live testing.

7.3 Future Recommendations

- Add a dedicated firewall at the internet edge.
- Configure port security on access ports to prevent MAC flooding.
- Deploy Syslog and SNMP for centralised monitoring.
- Add redundancy, such as a second core switch and STP tuning, for high availability.

7.4 Final Statement

The network is functional, secure and ready for deployment. All client requirements, including the boardroom constraint and Change Request 9, have been satisfied and verified.

8. References

Cisco Systems. (2023). *Cisco IOS IP Access List Commands*. Cisco Press.

Odom, W. (2020). *CCNA 200-301 Official Cert Guide, Volumes 1 and 2*. Cisco Press.

Cisco Networking Academy. (2023). *CCNA: Switching, Routing, and Wireless Essentials*.

IEEE. (2018). *IEEE 802.1Q - VLAN Tagging Standard*.